



Enumeration

Enumeration is defined as the process of extracting user names, machine names, network resources, shares and services from a system.

In this phase, the attacker creates an active connection to the system and performs directed queries to gain more information about the target.

The gathered information is used to identify the vulnerabilities or weak points in system security and tries to exploit in the System gaining phase.

Enumeration is the key to a Successful Penetration Tester.

Which Types of Details we find in Enumeration :

Network Resource and shares

Users and Groups

Routing tables

Auditing and Service settings

Machine names

Applications and banners

DNS details

Machine name

Dirctory enumerate

Nbtscan : NBTscan is a program for scanning IP networks for NetBIOS name information. For each responded host it lists IP address, NetBIOS computer name, logged-in user name and MAC address.

Command - #nbtscan -r (ip range)(subnetting) - nbtscan -r 192.168.1.1/24 - For Ip Range

Command - #nbtscan ip address - 192.168.10.9 - For single ip

smbmap : SMBMap allows users to enumerate samba share drives across an entire domain. List share, drives, drive permissions, share contents, upload/download functionality, file name etc. This tool intended is to simplify searching for potentially sensitive data across large networks.

Command - #smbmap -H <Host ip> - smbmap -H 192.168.10.9

Command - #smbmap -H <host ip> -d <domain name> -u <username> -p <password> -
smbmap -H 192.168.223.128 -d metasploitable -u msfadmin -p msfadmin

Nmap : Following Script attempts to detect if a Microsoft SMBv1 server is vulnerable to a remote code execution vulnerability.

Command - #nmap --script smb-vuln* -p 139,445 <host ip> -d - #nmap --script smb-vuln* -p 139,445 192.168.10.9 -d

Use -d if it doesn't work

SMTP enumeration can be performed in many ways. The easiest way to do this is by connecting to the SMTP service port of the target with telnet.

#telnet <host ip address> <port number>

#telnet 192.168.223.128 25

Use the VRFY command to check if users "user", "msfadmin" and "root" exist in this system.

#telnet <host ip address> <port number>

#telnet 192.168.223.128 25

#vrfy admin

#vrfy msfadmin

#vrfy user

Enum4linux : enum4linux tool allow us to extract usernames and more information from target.

Command - #enum4linux -a <ip address> - enum4linux -a 192.168.1.17

Metasploit :

Username enumeration through smtp using metasploit.

#msfconsole

#use auxiliary/scanner/smtp/smtp_enum

#auxiliary(smtp_enum) > set rhosts 192.168.1.107

#auxiliary(smtp_enum) > set rport 25

#auxiliary(smtp_enum) > set USER_FILE /root/Desktop/user.txt

#auxiliary(smtp_enum) > exploit

Dirb : We can use dirb tool for enumerating directories if http port 80 is open

Command - #dirb <ip address> - dirb http://192.168.1.17

**#dirb <ip address or domain name> <wordlist path> - dirb http://192.168.1.17
/root/usr/share/dirb/wordlists.big.txt**